EXPERIMENT-12

Z-SCOORE NORMALIZATION:

AIM:

To write the program for Z-scoore normalization using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")

age <- c(23, 24, 25)

marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

z_score <- scale(df$marks)

print(z_score)
```

RESULT:

Thus the Z-scoore normalization using R tool was executed successfully.

```
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> z_score <- scale(df$marks)
>
> print(z_score)
           [,1]
[1,]  0.7186975
[2,]  0.4233424
[3,] -1.1420399
attr(,"scaled:center")
[1] 63.66667
attr(,"scaled:scale")
[1] 33.85754
>
```

EXPERIMENT-13

MIN,MAX,MEAN,MINMAX:

AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

```
names <- c("siri", "mahi", "chiru")

age <- c(23, 24, 25)

marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

min_marks <- min(df$marks)

max_marks <- max(df$marks)

minmax <- (df$marks - min_marks) / (max_marks - min_marks)

print(min_marks)

print(max_marks)

print(minmax)
```

RESULT:

Thus the program for min,max,minmax,mean was executed successfully.

```
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> min_marks <- min(df$marks)
> max_marks <- max(df$marks)
>
> minmax <- (df$marks - min_marks) / (max_marks - min_marks)
>
> print(min_marks)
[1] 25
> print(max_marks)
[1] 88
> print(minmax)
[1] 1.0000000 0.8412698 0.0000000
```

EXPERIMENT-14

BAR PLOT AND HORIZONTAL BAR:

AIM:

To draw the bar plot and horizontal bar using R-tool.

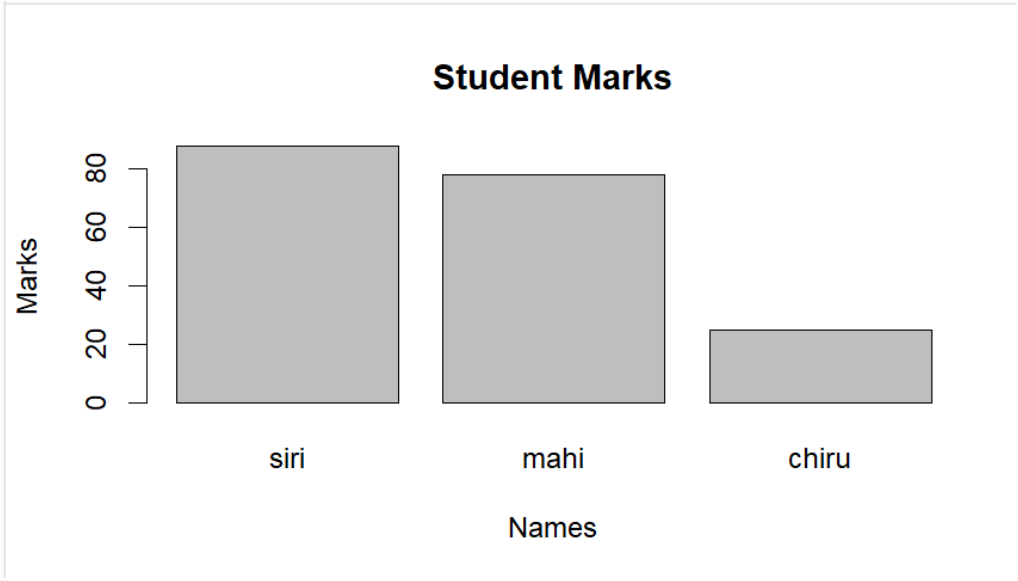
PROGRAM:

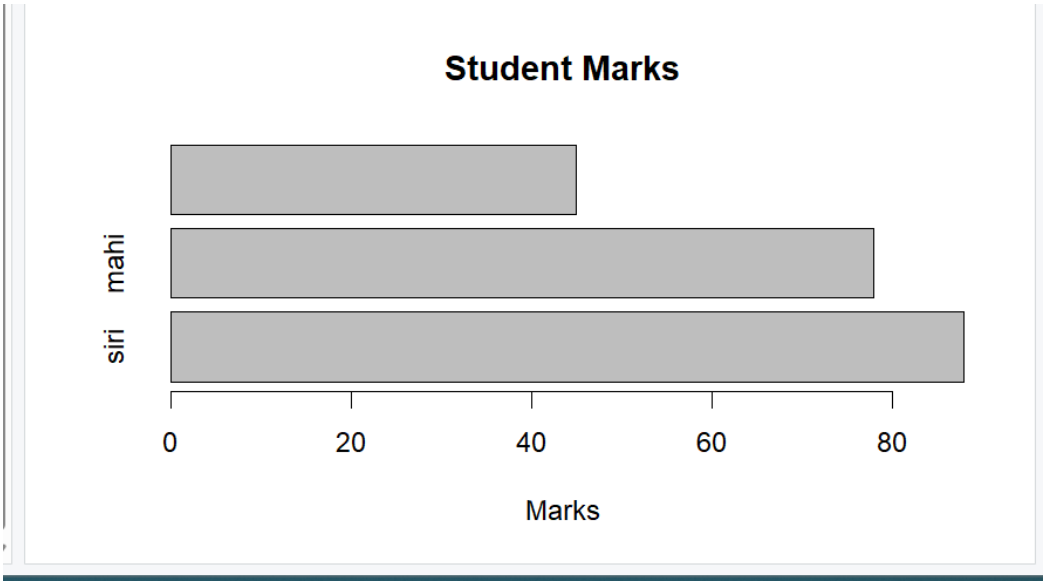
```
names <- c("siri", "mahi", "chiru")
marks <- c(88, 78, 25)
barplot(marks,
    names.arg = names,
    main = "Student Marks",
    xlab = "Names",
    ylab = "Marks")
names <- c("siri", "mahi", "chiru")
marks <- c(88, 78, 25)


barplot(marks,
    names.arg = names,
    main = "Student Marks",
    xlab = "Marks",
    horiz = TRUE)
```

RESULT:

Thus the bar and horizontal bar plot was executed successfully.


Student Marks
Marks
0
20
40
60
80
siri
mahi
chiru
Names


Student Marks
siri
mahi
0
20
40
60
80
Marks

EXPERIMENT-15

BOX PLOT:

AIM:

To draw the box plot using R-tool.

PROGRAM:

```
names<-c("siri","chru","loki")

age<-c(23,24,25)

marks<-c(88,78,25)

df<-data.frame(names,age,marks)

hist(df$age)

boxplot(df$age)
```

RESULT:

Thus the box plot was executed successfully.

